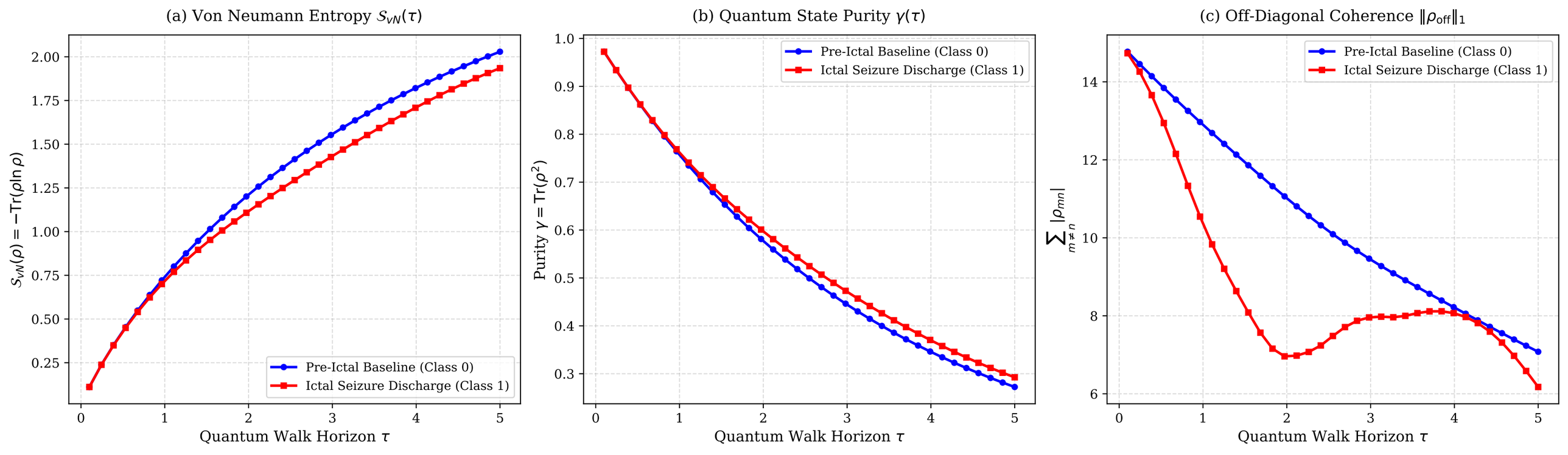
(a) Von Neumann Entropy $\mathcal{S}_{vN}(\tau)$
$\mathcal{S}_{vN}(\rho) = -\mathrm{Tr}(\rho \ln \rho)$
Quantum Walk Horizon τ
Pre-Ictal Baseline (Class 0)
Ictal Seizure Discharge (Class 1)
(b) Quantum State Purity $\gamma(\tau)$
Purity $\gamma = \mathrm{Tr}(\rho^2)$
Quantum Walk Horizon τ
Pre-Ictal Baseline (Class 0)
Ictal Seizure Discharge (Class 1)
(c) Off-Diagonal Coherence $\|\rho_{\mathrm{off}}\|_1$
$\sum_{m \neq n} |\rho_{mn}|$
Quantum Walk Horizon τ
Pre-Ictal Baseline (Class 0)
Ictal Seizure Discharge (Class 1)